

Copyright
by
Chenyu Tian
2025

The Dissertation Committee for Chenyu Tian
certifies that this is the approved version of the following dissertation:

Simulation of partial melting material

Committee:

Todd Arbogast, Supervisor

Marc Hesse, Co-supervisor

Omar Ghattas

Patric Heimbach

Thorsten Becker

Simulation of partial melting material

by
Chenyu Tian

Dissertation

Presented to the Faculty of the Graduate School of
The University of Texas at Austin
in Partial Fulfillment
of the Requirements
for the Degree of

Doctor of Philosophy

The University of Texas at Austin
May 2025

Dedication

To all future Longhorn graduate students.

Epigraph

What starts here changes the world.

—University of Texas at Austin

Acknowledgments

Thanks to the previous authors of this template, and to the Graduate School for providing it.

Preface

Readers, this document represents a great deal of time and effort. Enjoy.
Cheers, Chenyu Tian.

Abstract

Simulation of partial melting material

Chenyu Tian, PhD
The University of Texas at Austin, 2025

SUPERVISORS: Todd Arbogast, Marc Hesse

Table of Contents

List of Tables	10
List of Figures	11
Chapter 1: Governing Equations	12
1.1 Mechanics of Mantle Dynamics	12
1.1.1 Scaled variables	14
1.2 Conservation of Chemical Species and Energy	15
1.2.1 Effective Velocity	17
1.3 Eutectic Phase Behavior	18
1.3.1 Eutectic Phase Parameters	18
1.3.2 Eutectic Phase Split	19
1.3.3 Melting and Eutectic Temperature	20
1.4 Nondimensionalization	20
Works Cited	22

List of Tables

1.1 Parameters Table 21

List of Figures

1.1	An representative volume element (RVE).	16
1.2	Linear Eutectic Phase Splitting.	19

Chapter 1: Governing Equations in Mantle Dynamics

1.1 Mechanics of Mantle Dynamics

Fluid dynamics of the mixture of two phase flow has been extensively discussed. We will discuss the three governing equations of mass and momentum referring to the similar discussion in Drew and Segel (1971), McKenzie (1984) and Katz (2022). Considering a conservation equation for given volumetric variable γ in a representative volume element V .

$$\frac{d}{dt} \int_V \gamma d\mathbf{v} = - \int_{\partial V} \mathbf{f} \cdot d\sigma + \int_V S d\mathbf{v} \quad (1.1)$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V . The minus sign indicates that the positive directions aligns with normal unit vector pointing outwards. S , source/sink term, represents the volumetric production rate of γ . We can apply Gauss's theorem to obtain differential form,

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot \mathbf{f} = S \quad (1.2)$$

We also define the quantity of phase fraction rigorously with the help of an indicator function admitting that multiple phase cannot coexist at the same spatial point, which are regarded as immiscible phases. We define indicator function,

$$i(\mathbf{x}) = \begin{cases} 1 & \text{phase}_i \\ 0 & \text{not phase}_i \end{cases} \quad (1.3)$$

We then define macroscopic phase fraction function:

$$\phi_i = \frac{1}{\delta \mathbf{v}} \int_{\delta \mathbf{v}} i(\mathbf{x}) d\mathbf{v} \quad (1.4)$$

In the beginning of my discussion, We will discuss two immiscible phases liquid magma and solid mantle. Phase fraction of liquid magma is labelled as ϕ_l and phase fraction of solid mantle is labelled as ϕ_s , where $\phi_l + \phi_s = 1$ is always attained. Phase fraction

can also be regarded as volume fraction in macroscopic sense. Substituting bulk density $\bar{\rho} = \phi_l \rho_l + \phi_s \rho_s$ for generalized volumetric variable γ in equation 1.1, we can obtain conservation law for bulk mass,

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} = 0 \quad (1.5)$$

with the help of Boussinesq approximation ignoring compressibility of each individual phase, and extend Boussinesq approximation further applying to density difference between each phase except body force term. We can write continuous function for two phase mixture simply as,

$$\nabla \cdot \bar{\mathbf{v}} = 0 \quad (1.6)$$

Substituting phase averaged momentum $\bar{\rho} \bar{\mathbf{v}}$, We then write conservation of momentum with phase averaged term,

$$\frac{\partial \bar{\rho} \bar{\mathbf{v}}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} \otimes \mathbf{v} = \bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{P} \quad (1.7)$$

where $\boldsymbol{\tau}$ represents deviatoric stress, Ignoring change of inertia term due to near zero Reynolds number in magma and mantle,

$$\bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{P} = 0 \quad (1.8)$$

Insert more details later!

Combining equations, I finally arrive with the full flow mechanical system,

$$\phi_l (\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l} \nabla (p_l - p_f \mathbf{g}) = 0, \quad (1.9)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (1.10)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) = (\rho_s - \rho_l) \mathbf{g}, \quad (1.11)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_f (p_l - p_s) = 0, \quad (1.12)$$

where the deviatoric stress of the mixture is

$$\tau(\mathbf{v}_s) = 2\mu_s\phi_s(\mathcal{D}\mathbf{v}_s - \frac{1}{3}\nabla \cdot \mathbf{v}_s\mathbf{I}) \quad (1.13)$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla\mathbf{v}_s + \nabla\mathbf{v}_s^T)$ is the symmetric gradient. Moreover, the relative permeability is $k_{\phi_l} = k_0\phi_l^{2+2\Theta}$, where $\Theta \in [0, 1/2]$. μ_s and μ_l are dynamic viscosities for solid and liquid correspondingly.

1.1.1 Scaled variables

The Darcy part will mathematically degenerate in regions where the porosity is zero. To handle this issue, we use the formulation developed previously Arbogast et al. (2017). We first define pressure potentials as,

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \quad (1.14)$$

where $g = |\mathbf{g}|$ and z is the depth, so $\mathbf{g} = g\nabla z$. Note that ρ_f is used to define both potentials, so that with $q = \bar{q} = \bar{p} - \rho_l g z$,

$$p_l - p_s = q_l - q_s = \frac{1}{1 - \phi_l}(q_l - q).$$

We can then rewrite the system (1.9)–(1.12) omitting the subscript of liquid volumetric fraction,

$$\phi\mathbf{v}_r + \frac{k_0\phi^{2+2\Theta}}{\mu_l}\nabla q_l = 0, \quad (1.15)$$

$$\mu_s\nabla \cdot (\phi\mathbf{v}_r) + \frac{\phi}{1 - \phi}(q_l - q) = 0, \quad (1.16)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1 - \phi)\rho_r\mathbf{g}, \quad (1.17)$$

$$\mu_s\nabla \cdot \mathbf{v}_s - \frac{\phi}{1 - \phi}(q_l - q) = 0. \quad (1.18)$$

where relative velocity $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$ and density difference $\rho_r = \rho_s - \rho_l$. We refer to ϕ as porosity in the later text. Scaled variables are defined,

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta}\mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2}q_l, \quad (1.19)$$

and the problem is reformulated as

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla (\phi^{-1/2} \tilde{q}_l) = 0, \quad (1.20)$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (1.21)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1-\phi) \rho_r \mathbf{g}, \quad (1.22)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi} (\tilde{q}_l - \phi^{1/2} q) = 0. \quad (1.23)$$

Where the new scaled relative velocity maintain numerical stability in zero porosity region.

1.2 Conservation of Chemical Species and Energy

Chemical compositions of mantle and magma is very complicate and very hard to trace at the same time. We simplify the model to a two-phase system with two species. We negelect detailed description of chemical reaction process and effect on the Gibbs free energies. Similar discussion can be found in extensive literatures like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Marc A. Hesse and Oza (2022). Both two species can be solid or liquid. Two species are immiscible as solid but miscible as liquid. I consider a homogenous density ρ_s for solid phase and ρ_l for liquid phase. With extened Boussinesq approximation later assuming no density difference except in body force term, we can make further assumption that $\rho_l = \rho_s = \rho$ Within the representative volume element V , $\gamma_{i,j}$ represents one quantity of the specie i presenting in phase j , where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given specie i in phase j is often regarded as concentration. Hence, we can write transport equations for a individual specie i assuming a pseudo melting term Γ_i ,

$$\frac{\partial}{\partial t} (\phi_l \rho_l c_{i,l}) + \nabla \cdot (\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = \Gamma_i \quad (1.24)$$

$$\frac{\partial}{\partial t} (\phi_s \rho_s c_{i,s}) + \nabla \cdot (\phi_s \rho_s c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}) = -\Gamma_i \quad (1.25)$$

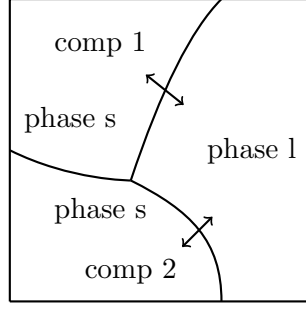


Figure 1.1: An representative volume element (RVE).

where we can safely get rid of dissuion term in solid phase. Summing up equation1.24 and equation1.25, we can write a advection-diffusion equation in phase averaged volumetric bulk variable,

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot (\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = 0 \quad (1.26)$$

where the phase averaged volumetric bulk variable are defined as $\overline{\rho c_i} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i \mathbf{v}} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted since only one chemical specie is traced in this two-component system. Extending Boussinessq approximation $\rho_l = \rho_s = \bar{\rho}$, the specie transport equation can be simplified as,

$$\frac{\partial \bar{c}}{\partial t} + \nabla \cdot (\bar{c} \mathbf{v} - \phi_l \mathcal{D}_l \nabla c_l) = 0 \quad (1.27)$$

We write energy balance equation in the phase-averaged internal energy form first,

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} - \nabla \cdot \overline{K \nabla T} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v} \cdot \mathbf{g}} \quad (1.28)$$

On the left hand side, we have time dependent rate of internal energy chage, advection term and diffusion term. On the right hand side, we have heating/cooling source $\mathcal{Q}$, work exerted by stress and work against gravity g . In a thermochemical equilibrium scenerio, we are replacing internal energy with enthalpy. We can write energy balancing equation neglecting additional heating/cooling term $\mathcal{Q}$ and dissipative heating McKenzie (1984),

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot \overline{\mathbf{v} T} = \rho L \nabla \cdot \phi_s \mathbf{v}_s + \overline{K \nabla^2 T} + \rho \alpha_\rho T \overline{\mathbf{v} \cdot \mathbf{g}} \quad (1.29)$$

where specific heat capacity c_p is used for both components in solid and liquid phase. Similar but slighted altered form was also discussed in Katz (2008), where adiabatic cooling effect was added in terms of potential temperature McKenzie and Bickle (1988). We now divide ρc_p on both sides of the equation considering Boussinesq approximation.

$$\frac{\partial H}{\partial t} + \nabla \cdot \bar{\mathbf{v}} T = \frac{L}{c_p} \nabla \cdot \phi_s \mathbf{v}_s + \kappa \nabla^2 T + \frac{\alpha_\rho}{c_p} T \bar{\mathbf{v}} \cdot \mathbf{g} \quad (1.30)$$

where $H = \bar{h}/c_p$ is the bulk specific enthalpy and $\kappa = \bar{K}/\rho c_p$ is the heat diffusivity.

1.2.1 Effective Velocity

Effective velocity in equilibrium transport for concentration within liquid phase is discussed in Spiegelman (1996). We also introduce our effective velocity, but instead of transporting concentration in liquid(magma), we transport bulk concentration. Define partition coefficient $D_i = c_{i,s}/c_{i,l}$ and

$$(\phi_l + \phi_s D_i) c_{i,l} = \bar{c}_i \quad (1.31)$$

$$(\phi_l/D_i + \phi_s) c_{i,s} = \bar{c}_i \quad (1.32)$$

Substitute $c_{i,l}$ and $c_{i,s}$ with equation 1.31 and equation 1.32 into bulk velocity $\bar{c}_i \bar{\mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l}$,

$$\bar{c}_i \bar{\mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i} \right) \bar{c}_i \quad (1.33)$$

Hence, we get the proper expression for the effective velocity as $\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}$, where the partition coefficient D can be obtained through phase behavior package. If there is no melting $\phi_l = 0$, we can retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. If there is all melting $\phi_l = 1$, we can obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Hence, the definition of effective velocity is well defined in those two boundary scenerios. We rewrite the transport of specie with effective velocity,

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0 \quad (1.34)$$

Where $C = \bar{c}$ represents volumetric phase averaged (bulk) concentration of species.

1.3 Eutectic Phase Behavior

The enthalpy method requires an explicit relationship between temperature and enthalpy. Embracing the fact that mid-ocean ridge forms at the place where temperature is generally lower, eutectic melting, where melting temperature is significantly lowered with the presence of the secondary specie, is a suitable model for modelling melting behavior at certain location. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution. On a typical eutectic phase behavior diagram, we can locate any phase situation using transported enthalpy and concentration. Binary eutectic melting has been applied to water-salt-brine system in glacier Marc A. Hesse and Oza (2022).

1.3.1 Eutectic Phase Parameters

In a eutectic system, we use bulk mass fraction and bulk enthalpy to locate the current state of a given system. $\mathcal{X}_i$ represents bulk mass fraction of specie i and $\mathcal{X}_{i,e}$ represents eutectic mass fraction of specie i in liquid. The subscript i will be omitted in the later context since only one specie is being traced in transport equation. Some assumptions are made for partially molten mantle system.

- Assuming chemical equilibrium is achieved that phase state can be calculated with phase diagram.
- The composition of the second specie in the liquid formed by eutectic melting $\mathcal{X}_e$ is constant at given pressure $\mathcal{X}_e(P)$.
- Melting temperature T_m is constant at given pressure $T_m(P)$.
- Boussinesq approximation is extended to phase behavior as well. assuming $\rho_l = \rho_s$.
- Assuming no density difference between two solid components.
- Assuming constant thermal expansion and specific heat capacity

We can now relate bulk mass fraction with bulk concentration directly,

$$\mathcal{X} = \frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = C = \phi_2 + \phi_l c_l \quad (1.35)$$

We relate bulk specific enthalpy with temperature and latent heat again.

$$H = T + \phi_l L / c_p \quad (1.36)$$

we further rescale by dividing melting temperature T_{m0} at standard atmospheric pressure forming a dimensionless relationship,

$$\check{H} = \check{T} + \phi_l \check{L} \quad (1.37)$$

where $\check{L} = \frac{L}{c_p T_{m0}}$ is the dimensionless latent heat.

1.3.2 Eutectic Phase Split

For simplicity, we are now assuming a linear eutectic phase split between different phase regions. We can draw the eutectic phase diagram separating different phase regions with straight lines at a fixed pressure. We now describe all six phase

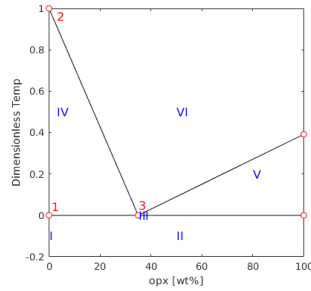


Figure 1.2: Linear Eutectic Phase Splitting.

regions depicting in the phase split diagram above,

I Single-phase sub-solidus : solid1

II Two-phase sub-solidus : solid1 + solid2

III Three-phase eutectic : solid1 + solid2 + melt

IV Two-phase super-eutectic : solid1 + melt

V Two-phase super-eutectic : solid2 + melt

VI Single-phase super-liquidus : melt

1.3.3 Melting and Eutectic Temperature

We approximate the melting temperature and eutectic temperature at a given pressure using Calusius-Clapeyron relationship,

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1}P \\ T_e &= T_{e0} + \nu^{-1}P \end{aligned} \tag{1.38}$$

where μ is the Clausius-Clapeyron constant.

1.4 Nondimensionalization

Compaction and melting are two most important aspects in mantle dynamics. We nondimensionlize the scaled formulation with characteristic length scale related to compaction length and scaled characteristic Darcy speed of the melt,

$$l_0 = \left(\frac{k_0 \mu_s}{\mu_l} \right)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0 \rho_r g}{\mu_l} \tag{1.39}$$

We can derive corresponding characteristic time scale as $t_0 = l_0/u_0$, now we can write full set of dimensionless equations. We represent dimensionless variables with a $\checkmark$ mark,

$$\check{\mathbf{v}}_r + \phi^{1+\Theta} \nabla (\phi^{-1/2} \check{q}_l) = 0, \tag{1.40}$$

$$\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1-\phi} (\check{q}_l - \phi^{1/2} \check{q}) = 0, \tag{1.41}$$

$$\nabla \check{q} - \nabla \cdot \check{\boldsymbol{\tau}}(\check{\mathbf{v}}_s) = -(1-\phi) \mathbf{n}, \tag{1.42}$$

$$\nabla \cdot \check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi} (\check{q}_l - \phi^{1/2} \check{q}) = 0. \tag{1.43}$$

Quantity	Value	Units
ρ	3000	$kg \cdot m^{-3}$
ρ_r	600	$kg \cdot m^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$J \cdot kg^{-1} \cdot K^{-1}$
L	5×10^5	$J \cdot kg^{-1}$
μ_l	1	Pascal $\cdot s$
μ_s	10^{19}	Pascal $\cdot s$
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	Pascal $\cdot K^{-1}$

Table 1.1: Flow Mechanics and Thermodynamics Parameters

where dimensionless deviatoric stress of the mixture is $\check{\tau} = 2(1 - \phi)(\mathcal{D}\check{\mathbf{v}}_s - 1/3\nabla \cdot \check{\mathbf{v}}_s \mathbf{I})$ and $\mathbf{n}$ is the unit direction vector pointing downwards. Now we align transport equations with dimensionless variables defined in mechanic system,

$$\begin{aligned}
\frac{\partial C}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}_{\text{eff}} C - \phi \mathcal{D}_l / u_0 \nabla c_l) &= 0 \\
\frac{\partial \check{H}}{\partial \check{t}} + \nabla \cdot \check{\mathbf{v}} T - \check{L} \nabla \cdot \phi_s \check{\mathbf{v}}_s - \frac{\kappa}{u_0 l_0} \nabla^2 \check{T} - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\nabla} \cdot \mathbf{g} &= 0
\end{aligned} \tag{1.44}$$

Parameters used in the flow mechanics and thermodynamic computations are listed in Table 1.1,

Works Cited

- T. Arbogast, M. A. Hesse, and A. L. Taicher. Mixed methods for two-phase Darcy-Stokes mixtures of partially melted materials with regions of zero porosity. *SIAM J. Sci. Comput.*, 39(2):B375–B402, 2017. DOI 10.1137/16M1091095.
- Donald Drew and Lee Segel. Averaged equations for two-phase flows. *Studies in Applied Mathematics*, page 3, 1971.
- I.J Hewitt and A.C Fowler. Partial melting in an upwelling mantle column. *Proc. R. Soc. A*, 464:2467–2491, 2008.
- Richard F. Katz. Magma dynamics with the enthalpy method: Benchmark solutions and magmatic focusing at mid-ocean ridges. *Journal of Petrology*, 49: 2099–2121, 2008.
- Richard F. Katz. *The Dynamics of Partially Molten Rock*. Princeton University Press, 1st edition, 2022. ISBN 9780691232645.
- Steve D. Vance Marc A. Hesse, Jacob S. Jordan and Apurva V. Oza. Supporting information for ”downward oxidant transport through europa’s ice shell by density-driven brine percolation”. *Geophysical Research Letters*, 2022.
- Dan Mckenzie. The generation and compaction of partially molten rock. *Journal of Petrology*, 25:713–765, 1984.
- Dan Mckenzie and Mike J. Bickle. The volume and composition of melt generated by extension of the lithosphere. *Journal of Petrology*, 29:625–679, 1988.
- Marc Spiegelman. Geochemical consequences of melt transport in 2-d: The sensitivity of trace elements to mantle dynamics. *Earth and Planetary Science Letters*, pages 115–132, 1996.

Marc Spiegelman and Tim Elliott. Consequences of melt transport for uranium series disequilibrium in young lavas. *Earth and Planetary Science Letters*, 118: 1–20, 1993.